DOROTHY HODGKIN

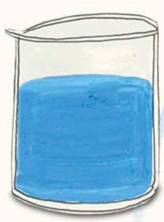
British chemist
1910–1994

Dorothy Hodgkin couldn't wait until she was grown up to become a scientist. She turned the attic of her house into a lab, collecting objects she found in nature and analyzing them with her chemistry set. However, what captured Dorothy's attention most were the beautiful crystals that formed when certain salts were dissolved in water. She decided to study chemistry in college, so she could keep exploring.

Dorothy learned a new way to unlock the secrets of chemicals. X-rays could be used to peer deep inside the crystal of a chemical, to figure out what it was made from. Dorothy focused on chemicals that were important to human health.

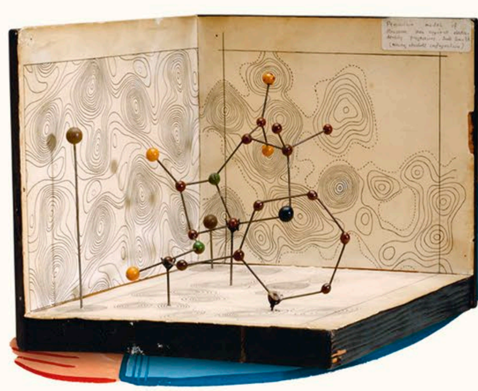
To get hold of crystals to examine, Dorothy used a version of the simple experiments she did in school.

1) Dissolve lots of copper sulfate in water.



2) The water slowly evaporates (turns to gas).

3) What's left behind is pure copper sulfate. Its molecules have a regular shape, so they pack together neatly to form crystals.



Dorothy's first big discovery was the structure of the antibiotic penicillin. Scientists could now make their own versions of this important medicine.